4	Two parallel plates are in a vacuum. One plate is positively charged and the other plate is earthed. A rectangular conductor of width x is placed in between the plates so that one of its faces is at a distance $0.5x$ from the positively charged plate and the opposite face is at $1.5x$ from the earthed plate as shown in Fig. 4.1.

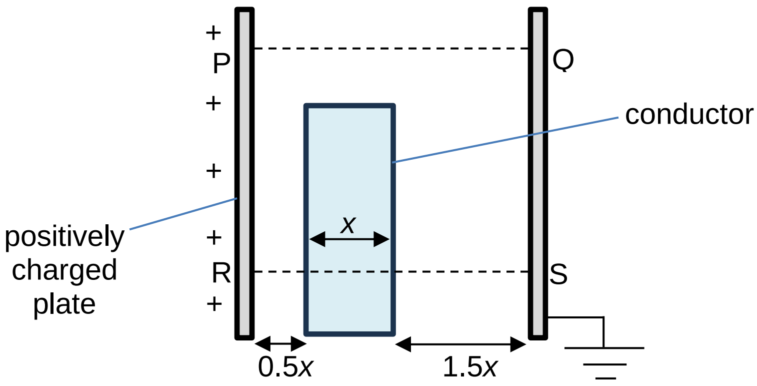


Fig. 4.1

The electric potential difference across the parallel plates is 3.00 V.

(a) The variation with distance from P to Q of the electric potential along line PQ is shown in Fig. 4.2.

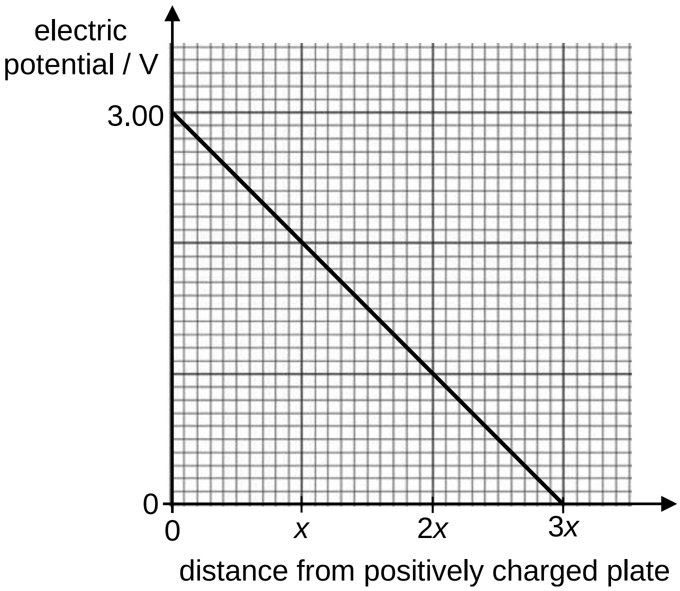
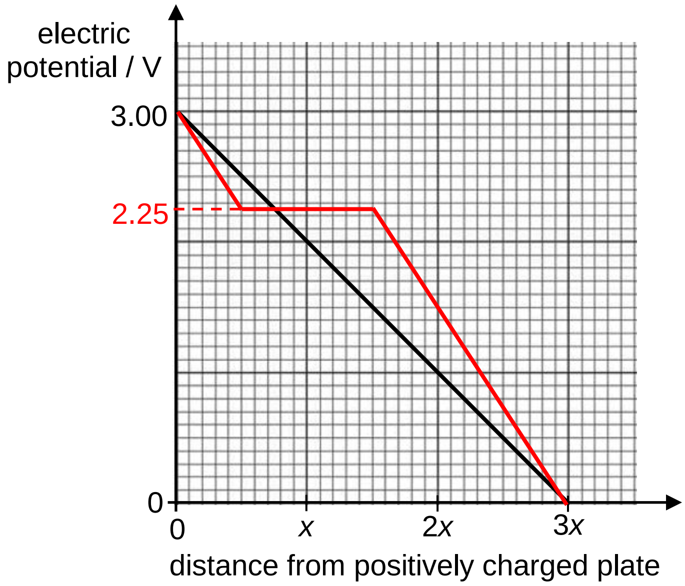


Fig. 4.2

On Fig. 4.2, draw a line to show the variation with distance from R to S of the electric potential along the line RS. [2]

L3 The electric potential **decreases proportionally** with distance in **between parallel plates**.
In addition, there is **no change in electric potential across the conductor**.
Therefore,



B1
B1

1 mark – Horizontal line (at 2.25 V) **from 0.5*x* to 1.5*x*.**
1 mark – Straight line from 3.00 V to 2.25 V on the left of the conductor AND straight line from 2.25 V to 0 V to the right of the conductor, AND **both** of these lines must have **equal** gradients, i.e. equal electric field strength.

Working:
Since:
- Same uniform electric field strength outside the conductor, and,
- Zero p.d. within the conductor, and,
- Total p.d. between parallel plates equal to 3.00 V.

Thus:
- Effective distance over which potential decreases by 3.00 V is $2x$ (= plate separation – width of conductor),
- Electric field strength outside the conductor, $E_{outsdie} = 3.00 / 2x = 1.5x^{-1}$,
- To the left of the conductor, p.d. = $E_{outsdie}.0.5x = (1.5x^{-1})(0.5x) = 0.75$ V → potential of left face of conductor = 3.00 – 0.75 = 2.25 V.

(b) The variation with distance from P to Q of the electric field strength along line PQ is shown in Fig. 4.3.

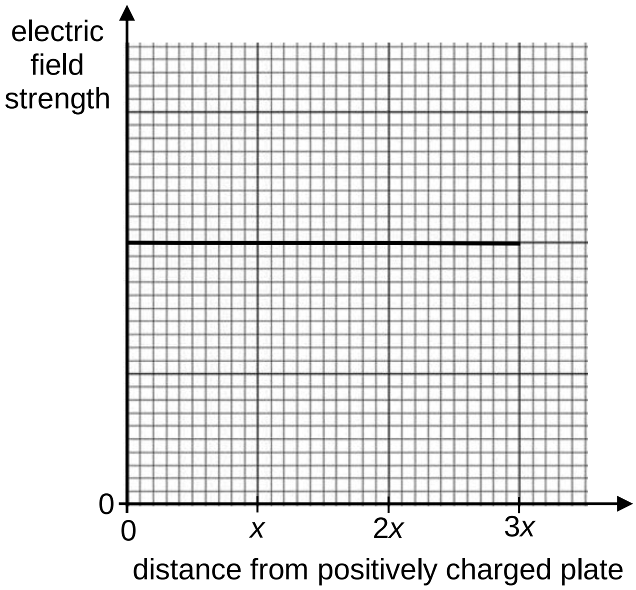


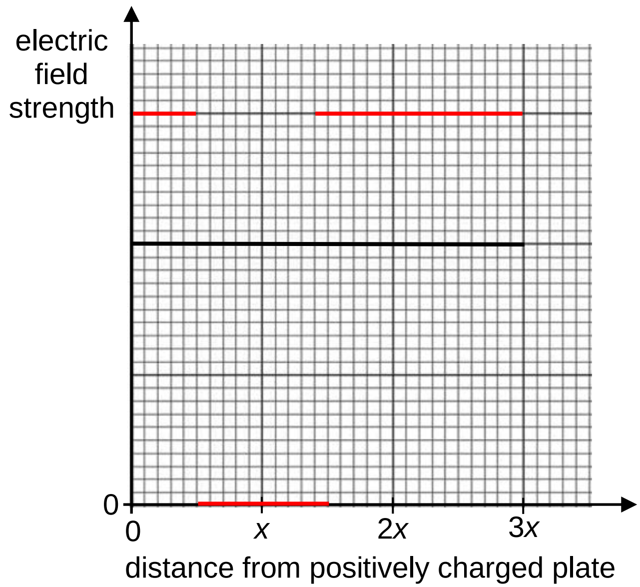
Fig. 4.3

On Fig. 4.3, draw a line to show the variation with distance from R to S of the electric field strength along the line RS. [3]

L3

The electric field strength is assumed to be constant in between parallel plates, and the electric field strength in a conductor is zero.

By observing the gradient of the graph in Fig. 4.2, the electric field strength is the same between 0 to 0.5*x* and from 1.5*x* to 3*x*.



1 mark – Horizontal straight line at **0** between **0.5*x* to 1.5*x*** **B1**

1 mark – Horizontal, **non-zero** lines to depict constant non-zero electric field strength between **0 to 0.5 *x*** and **1.5*x* to 3*x*, and at greater electric field strength** **B1**

1 mark – Magnitude of electric field strength outside the conductor is at **1.5 times** the value in PQ, and, the same between 0 to 0.5 *x* and 1.5*x* to 3*x* **B1**

Allow ECF from graph in part (a), showing understanding of E = -dV/dx.

Note to students: Phrasing of "draw a line..." follows A-level phrasing in 2020 P2 Q3, although one may argue that it is misleading since final answer has 3 separated lines not a single continuous line.

[Total: 5]

5 Fig. 5.1 shows a circular coil of 500 turns and radius 0.12 m.